

High Frequency Traders and Market Structure¹

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Abstract

The arrival of high-frequency traders (HFTs) coincided with the entry of new markets and, subsequently, strong fragmentation of the order flow. These trends might be related as new markets serve HFTs who seek low fees and high speed. New markets only thrive on competitive price quotes which well-connected HFTs can deliver as they can offload any nonzero position in any market they are connected to. HFTs may benefit or hurt market quality through adverse selection on price quotes, a technology arms race, or high-risk trading strategies.

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1 Introduction

Securities' trading is a search process to transfer securities for which buyer (expected) utility from owning the asset exceeds seller utility, due, perhaps, from differing private values (e.g., idiosyncratic hedge values). The social value produced is then referred to as gains-from-trade. If this differential is due to information on the security's common value (e.g., the mean of cash flow), then the transaction price most likely reveals information. Such information production is useful in allocating scarce resources in an economy (for reviews of the microstructure literature see e.g., O'Hara, 1995; Madhavan, 2000; or Foucault, Pagano and Röell, 2013).

In a pre-electronic age, the search process was particularly costly. A single centralized market could capitalize on a large network externality, creating a high entry barrier for potential rival markets. This externality has changed in the electronic age as search cost has become close to trivial. Because computers can easily check prices in a potential rival market, the network externality is substantially reduced and the incumbent market becomes contestable (see Domowitz and Steil, 1999). New markets get started and investors benefit from competition in terms of lower fees and innovation, but they also benefit from market heterogeneity (see Stoll, 2001).

Computers are used to match orders in an electronic market but they are also useful for implementing automated trading strategies, a practice referred to as 'algorithmic trading' (see Hendershott, Jones and Menkveld, 2011). There are two types of algorithms: agency algorithms, which implement a long-term position change at the lowest possible cost, and proprietary algorithms, which trade to lock in a trading profit and positions are relatively short-lived. Agents who run such algorithms are commonly referred to as high-frequency traders (HFT). As a visual guide, Figure 1 provides a schematic of gains-to-trade (Panel A) and the changes in the institutional trading structures (Panel B).

[insert Figure 1 about here]

2 Recent trends, market fragmentation, and HFT

One salient feature of equity trading in the 2000s is the sharp increase in quote activity. Angel, Harris and Spatt (2011) show that the average number of quotes per minute per stock has increased from 20 to 200 with even higher levels reached during the 2007-2008 financial crisis. Although this increase could reflect an increase in the number of trades, it would only partially explain the quote surge as Angel et al. further document that the cancel-to-trade ratio increased from 10 to 30 in the 2000s. The lower increase in this ratio relative to the quote activity intensity shows that the activity increase is both due to more trades and more quote cancellations in between trades. Overall, quote activity has increased sharply in the 2000s, more so than trade activity.

[insert Figure 2 about here]

Figure 2 illustrates that both in Europe and in the U.S. entrant markets have captured a significant market share towards the end of the decade (see also O'Hara and Ye, 2011). In Europe, the new entrants are Chi-X, BATS, and Turquoise who collectively grew from a zero market share at the start of 2007 to a 20% share in January 2011. In the U.S., the most striking example is that NYSE market share in NYSE-listed stocks fell from 80% in January 2003 to about 25% in January 2011. In 2013, BATS merged with Direct Edge. The merger created an exchange that matches the NYSE in terms of volume (see Bunge, 2013). In general, new markets obtained significant market share in the second half of the 2000s, both in the U.S. and in Europe.

At about the time that the new platforms made significant inroads into securities trading, proprietary algorithms operated by high-frequency traders (HFT) started to flourish. Their participation in trades is often estimated to be higher than 50%. For example, Brogaard (2010) documents a participation rate of 73.7% for a 2010 sample of NASDAQ stocks. Larry Tabb, chief executive of Tabb Group, a consultancy, said high frequency trading accounted for 54% of U.S. equity trading and 35% of European equity trading (see Grant, 2011). It appears that high-frequency traders' (i.e., proprietary algorithms)

participation in trades was small in the middle of the 2000s but, reportedly, grew to 30-70% by the end of the decade.

While agency algorithms are used to establish a long-term position change at the lowest possible transaction cost (by, for example, splitting a large order optimally across markets and across time), HFTs (and proprietary algorithms more generally) are characterized as automated trading strategies implemented on lightning-fast computers that operate at the shortest possible distance from the matching engine of a market. The objective of HFTs and proprietary algorithms is to lock in a trading profit. HFTs typically operate on intraday frequencies; they start and end the day on a zero net position (see SEC (2010a) for a detailed HFT characterization).

[insert Figure 3 about here]

Figure 3, which is taken from Jovanovic and Menkveld (2011), illustrates the trading of an HFT in a Dutch stock. It trades the stock in two markets, the incumbent market Euronext and the entrant market Chi-X. The HFT starts on a zero position, trades thousands of shares throughout the day, and ends on a zero position. The HFT is clearly not engaging in plain-vanilla arbitrage across the two platforms as that involves buying and *simultaneously* selling in the other. If this were the case, the HFT position should be zero throughout the day, whereas the graph shows that its position ranges between short 100,000 shares and long 100,000 shares. A nonzero position seems to last up to 30 minutes.

[insert Figure 4 about here]

Menkveld (2013) suggests that the success of the new platform Chi-X and the surge in high frequency trading are related. Figure 4 depicts the entrant market share in Dutch stocks as of the day of its inception, April 16, 2007. Interestingly, market share hovered around 1% in the early months of operation until, suddenly, in August it grew to a double-digit share. The same graph plots the overall market share of a new ‘broker’ that operates in both Chi-X and Euronext. It first appeared in the sample in August of that year. By the end of the month it participated in 10-15% of all trades and about half of its trades

materialized in Chi-X and the other half in Euronext. This broker's trading fits the SEC characterization of an HFT.¹

This evidence on the start of Chi-X supports the vision that was illustrated in Figure 1. A new market successfully enters because high-frequency traders can post bid and ask quotes and, if hit, they use both the entrant and the incumbent market to lay off the position. In this example, the entrant market was superior to the incumbent market as its fees were lower and it ran a faster system. These features are particularly important for a high-frequency trader who, by the nature of his operation, pays the fee (or earns the rebate) frequently and cares about speed in order to change its price quotes quickly on incoming information. This example supports the idea that entrant markets benefit from the presence of high-frequency traders and high-frequency traders thrive on the competition between markets. There appears to be a symbiotic relationship between HFT and new market entry.

To continue with this example, the competition for order flow that started with Chi-X entry had an immediate effect on both market quality and trading fees for investors at large. Euronext responded to the fast new system by significantly upgrading the speed of its own system. It lowered fees to reduce the gap with the fees charged by the entrant market. The strongest effect, however, was the reduction in clearing fees. With the entry of Chi-X a new clearing house entered, EMCF, and its aggressive fee structure triggered a fee war with Euronext's LCH-Clearnet. Within a year fees were 50% lower, which created substantial cost savings for market participants as clearing fees are the same order of magnitude as the fees charged by exchanges. In addition to these explicit trading cost reductions, the bottom graph of Figure 4 shows that also implicit cost declined as the bid-ask spread was reduced by about 30%.

The migration to electronic trading coincided with a gradual decline in the average transaction cost but it also coincided with the occasional disappearance of liquidity supply. The data service firm

¹ The high frequency trader whose arrival appears intimately linked to the success of Chi-X can be characterized as an electronic market maker. In most of its trades the HFT is on the passive side i.e., hers is the limit order that finds execution on an incoming market order. A decomposition of this HFT's profit shows that it makes money from these passive orders (through rebates and earning the spread) but loses money on the inventory positions it carries through time. Further details on the nature, profitability, and capital intensity of the HFT operation can be found in Menkveld (2013).

Nanex maintains an archive of equity market ‘mini crashes’ on their website.² The most dramatic example is the Flash Crash of May 6, 2010, when a large fundamental seller, reportedly, demanded substantial liquidity by trying to sell \$4.1 billion worth of the E-mini futures contract on the S&P 500. He initiated the sell order at 2:32 p.m. through an agency algorithm that was set to work the order aggressively as it, reportedly, was not restricted by a price limit (see SEC, 2010*b*). By 2:45 p.m. the price decline relative to the start of the day was 8% and most of it occurred after the initiation of the sell order. Not only were E-Mini prices affected, the sell order also caused similar price declines in related index-based securities, and even larger declines in some of the securities in the index (see SEC, 2010*b*).³

A natural reason for, at times, shallow liquidity supply is that HFTs avoid large positions that might take long to unwind. Such positions are inherently costly as they carry price risk but also, to the extent that they are carried overnight, tie up capital due to end-of-day margin requirements. A further reason for erratic supply is that machines might withdraw from the market if they observe price paths that they do not ‘understand’ e.g., a path that does not match any historical pattern so that the machine does not have an informational (predictive) advantage. In a traditional floor market, the exchange often designated a market maker who had the affirmative obligation to ‘lean against the wind,’ in particular at times when others withdraw from liquidity supply. HFTs are under no such obligation. Thus, it appears that costs seem to feature more frequent and more extreme spikes when migrating to electronic trading, with the Flash Crash as just one, albeit dramatic, example.

² See http://www.nanex.net/FlashCrashEquities/FlashCrashAnalysis_Equities.html.

³ Menkveld and Yueshen (2012) analyze a proprietary dataset that contains the order flow of the large seller. They document that in the two-minute period before the halt, when the market crashed, the seller’s order flow did not have immediate price impact. Instead, it seemed to have triggered, with a 300 millisecond delay, disproportionate aggressive selling by others.

3 Looking forward: the challenges, costs, and benefits of electronic trading

Does the presence of high-frequency traders affect investors' welfare at a more fundamental level than through producing competition between markets? Arguably an important advantage of machines over humans is their ability to process vast amounts of information at superhuman speed. One source of information is price quotes in for example index futures, same industry stocks, oil, or fixed income. Another source is textual analysis of news flow (see Gross-Klussmann and Hautsch, 2011). Perhaps the best example of their focus on an informational speed advantage is that, reportedly, at the Chicago Mercantile Exchange (CME), HFTs use limit orders to receive market information before others. As soon as an incoming market order hits their quote, the confirmation message is sent to the HFT before the trade information is publicly disseminated (see Patterson, Strasburg, and Plevin, 2013). In sum, HFT strategies are therefore likely to interact with the information asymmetry friction in securities trading. This friction and its consequence is formally analyzed in Jovanovic and Menkveld, 2011). A simple example illustrates their results:

Let us consider a market with investor arrivals that are spaced out in time (cf. Grossman and Miller, 1988) and, for simplicity, consider two investors: an early-arriving investor who owns an asset and puts it up for sale and a late-arriving investor who considers buying it. The seller posts an ask quote and leaves the market. The buyer arrives and will only buy the asset if her valuation for it is higher than the price posted. An informational friction arises when in between their arrivals some information is revealed publicly on the (common value of) the security. Naturally, only the potential buyer, the late investor, is aware of this information. If the information was good, the buyer is more inclined to take the asset as the (unaltered) price quote now appears relatively low. If it was bad, the seller is more likely to be stuck with the asset. The seller is therefore 'adversely selected' by the buyer. Ahead of time, the seller is aware that this might happen and therefore raises her price to protect herself. As a result, gains-from-trade are less likely to be realized.

The presence of competitive high-frequency traders with their ability to track the inter-arrival information can restore some of the gains-from-trade. One way to generate such effect is that high-frequency traders bid for the asset at the time the seller is in the market. Price competition (Bertrand) ensures that the HFT bid price is such that *ex-post* the HFT earns zero rent. The benefit of passing the asset off to the HFT is that machines have the ability to update their quote quickly⁴ on incoming information in which case *their* ask quote is not adversely selected by the buyer. This fact could explain why both the number of trades and the number of quotes per trade have increased: The number of trades increases as it becomes more likely that the seller transfers the asset to the buyer with a potential additional trade if it reaches the buyer via the HFT, while the number of quotes per trade increases as the HFT updates its ask quote on incoming information.⁵

The presence of high-frequency traders might, however, lower the gains-from-trade if the late investor, the buyer, was not aware of any inter-arrival information. In this case, the market did not suffer an informational friction to begin with. Entry of HFTs now generates adverse selection cost for the seller as the machine will snap up the asset on positive news. The seller rationally protects herself by raising the ask quote which, in effect, reduces the likelihood of trade. There is therefore less scope to realize gains-from-trade.

One might argue that in this case HFT entry reduces realized gains-from-trade, but still benefits the market in the sense that information gets impounded into prices more quickly i.e., the market becomes more efficient. The concern is whether this type of informational efficiency is socially productive in particular when such information would have become available to investors at lower frequencies. For example, an actively traded stock produces a couple of transactions per minute. The inter-arrival time of investors is therefore in the order of seconds. In such time frame there arguably will be price changes in

⁴ Tabb Group, a firm that tracks trends in electronic trading, estimates that 95% to 98% of orders submitted by HFTs are canceled (see Patterson and Ackerman, 2012).

⁵ This benevolent interpretation contrasts with the quote-stuffing interpretation, which explains bursts of quote activity as a way for a machine to deliberately generate congestion, thereby delaying others, and benefit itself (see Egginton, Van Ness and Van Ness, 2013; Gai, Yao and Ye, 2013).

the index futures market of which a human investor might not be aware. HFTs will use such information to ‘pick off’ human quotes with a delay that is within milliseconds (see McNish and Upson, 2013).⁶ The ‘clock speed’ of humans on a trading floor is much lower.⁷ Blume and Goldstein (1997) for example report a median delay of 16 seconds for quote updates after a trade at the NYSE floor market in 1990; McNish and Upson (2013) find that NBBO quote updates follow NYSE trades with a lag of 75 milliseconds in 2008. The larger point is that it is not the type of information that analysts push into stock prices after interviewing a company’s new management team, learn about their quality, and recommend buy, sell, or hold. Such information would not have been produced in the absence of analysts (or it would become available at such low frequency e.g., after quarters of disappointing sales, that the damage has already been done).⁸

Therefore, the presence of high-frequency traders in electronic markets improves welfare when they act as market makers and thereby reduce the informational friction that exists between non-synchronously arriving investors. The presence of high-frequency traders, however, reduces welfare when HFTs pick off investors’ quotes at superhuman speed on information that would have been revealed to investors anyway at a lower frequency.

A high-frequency trader often pays a venue to “co-locate” its trading robot close to the matching engine. In other words, the HFT pays to have a ‘fast door’ to the market. Such privilege might or might not be good for the market as a whole. To gauge its effect it is useful to condition on one of the two channels through which HFTs affect investor welfare. If an HFT uses the fast door to adjust its quotes on incoming information to avoid adverse selection, it most likely is good for investor welfare. If however,

⁶ McNish and Upson (2013) report that in 2008 ‘fast traders’ could earn up to \$233 million on U.S. equity exchanges merely because they can trade at a sub-second speed.

⁷ Overdahl and McMillan (1998) study index arbitrage activity directly based on a data sample from the late eighties through the early nineties. They show that index arbitrage reduces price discrepancies by up to 15% each minute.

⁸ The evidence mostly supports that HFT improves liquidity, replicates the role of market makers, and improves price efficiency (see Chaboud, Chiquoine, Hjalmarsson and Vega, 2014; Hendershott and Riordan, 2009; Brogaard, 2010; Hasbrouck and Saar, 2013; Hendershott, Jones and Menkveld, 2011; Menkveld, 2013).

an HFT uses it to aggressively pick-off other investors' quotes that are slow to change as investors travel through the main door, it is likely to destroy welfare.

Furthermore, high-frequency traders might be engaged in a socially wasteful arms race (see Biais, Foucault and Moinas, 2011). If one HFT invests in hardware or software to be faster to trade on an informational signal before a rival can, nothing is gained in terms of social value. True, the information is revealed more quickly but at a time scale that is unlikely to yield improved allocations. More importantly, a single HFT does not internalize the relative speed disadvantage it creates for others i.e., HFT technology investment is the source of a negative externality. As a result HFTs over-invest in technology relative to a social optimum (assuming there was no underinvestment in the absence of HFT). For example, Laughlin, Aguirre and Grundfest (2014) document a three-millisecond decrease in one-way communication between Chicago and New York in the period 2010-2012. They relate it to HFT investment into a new fiber optic cable and the introduction of line-of-sight microwave networks that, collectively, are estimated to cost more than \$500 million. For society at large, however, improved computing and communications technology might have social value if it is used to, for example, produce better medical equipment or an improved gaming experience. Overall, high-frequency traders over-invest in technology relative to a social optimum if the main motivation is to be ahead of rival HFTs when trading on a publicly observed signal.

One natural way to avoid a wasteful arms race is to redesign the standard continuous double auction market into a high-frequency series of batch auctions. The human eye is operating at a latency (response time) of 100 milliseconds as has been established experimentally based on a monkey's eye response time to a visual stimulus (see Krauzlis and Lisberger, 1994). If the auctions are scheduled at 10 Hertz (i.e., 10 auctions per second) it would enable humans to, at least, see the transaction price in the last auction before the next auction is due. Note that this does not annihilate a microsecond speed advantage as, if news events happen randomly, one still benefits from the ability to update a price quote on news that arrives one microsecond ahead of the next auction. The advantage is, however, substantially reduced as it pertains only to the small subset of news events that occur just before the next scheduled auction time. In

this example, it only pertains to 1 out of 100,000 news events if the latency competition is in the order of one microsecond (i.e., 100 milliseconds divided by one microsecond). Such a change would alleviate a serious cost to society due to the over-investment of HFT in technology used solely for the purpose of jumping ahead of rivals.

The speed of electronic trading also requires real-time monitoring of positions to ensure market integrity. For example, the environment makes the well-known ‘doubling strategy’ increasingly appealing. In roulette, such sure-win strategy is: bet on black, and keep doubling your bets until black is the outcome (see Harrison and Kreps, 1979). The strategy might require access to (infinitely) large credit and (infinitely) many trading rounds in case the player is unlucky in the early rounds. These conditions are increasingly met when markets operate at high frequency and risk controls operate at considerably lower frequency (e.g., some clearing houses base their margin calls on end-of-day positions with ad-hoc intraday monitoring).

A high-frequency trader might implement such doubling strategy by, for example, at the start of the day betting that sometime in the day the price will be higher than the opening price i.e., a bet on ‘black’. It starts the day with buying an initial position and keeps doubling it on falling prices until the market ‘hits black’. It unwinds the position at that price and makes a profit. Admittedly, such strategy requires negligible market impact when trading but for a thick enough market and a smart enough execution this might be the case. The strategy earns a ‘risk-free’ profit until, one day, doubling might be required so often and time might run out before the market hits black. In this case, the HFT will not have sufficient funds to meet the end-of-day margin call and the central clearing house (and therefore all other participants and/or the government) will have to take the loss. Therefore, in addition to the arms-race issue, high frequency electronic markets with low frequency system-wide monitoring potentially breed destructive doubling strategies.

4 Summary and conclusion

The arrival of high frequency traders and the strong fragmentation of markets are two recent phenomena that are tightly related. Entrant markets benefit from quotes provided by HFTs and HFTs benefit from the arrival of new low-fee and fast venues. The cost and benefit of HFT and fragmentation should therefore be considered jointly. One important benefit that HFT participation could have is that HFT participation might lower adverse-selection costs, as is the case when HFTs operate as market makers. HFTs might, however, raise such costs if they aggressively pick off the price quotes of others. Moreover, their participation could be socially costly when accompanied by a technology arms race or high-risk trading strategies e.g, the doubling strategy.

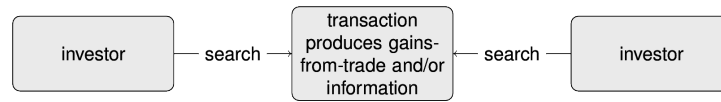
References

- Angel, James J., Lawrence E. Harris, and Chester S. Spatt, 2011. Equity trading in the 21st century. *Quarterly Journal of Finance* 1, 1–53.
- Biais, Bruno, Thierry Foucault, and Sophie Moinas, 2011. Equilibrium algorithmic trading. *Working paper*, Toulouse University.
- Blume, Marshall E., and Michael Goldstein, 1997. Quotes, order flow, and price discovery. *Journal of Finance* 52, 221–244.
- Brogaard, Jonathan A., 2010. High-frequency trading and its impact on market quality. *Working paper*, Kellogg School of Management.
- Bunge, Jacob, 2013. BATS prepares to take on big bell ringers, *Wall Street Journal*, September 27.
- Chaboud, Alain, Benjamin Chiquoine, Erik Hjalmarsson, and Clara Vega, 2014. Rise of the machines: Algorithmic trading in the foreign exchange market. *Journal of Finance*, forthcoming.
- Domowitz, Ian, and Benn Steil, 1999. Automation, trading costs, and the structure of the trading services industry. *Brooking-Wharton Papers on Financial Services*.

- Egginton, Jared, Bonnie F. Van Ness, and Robert A. Van Ness, 2013. Quote stuffing. *Working paper*, University of Mississippi.
- Foucault, Thierry, Marco Pagano, and Ailsa Röell, 2013. *Market Liquidity: Theory, Evidence, and Policy*. (Oxford University Press, Oxford).
- Gai, Jiading, Chen Yao, and Mao Ye, 2013. The externalities of high-frequency trading. *Working paper*, University of Illinois at Urbana-Champaign.
- Grant, Jeremy, 2011. High-frequency boom time hits slowdown, *Financial Times*, April 12.
- Gross-Klussmann, Axel and Nikolaus Hautsch, 2011. When machines read the news: Using automated text analytics to quantify high frequency news-implied market reactions. *Journal of Empirical Finance* 18, 321–340.
- Grossman, Sanford, and Merton H. Miller, 1988. Liquidity and market structure. *Journal of Finance* 43, 617–633.
- Harrison, J. Michael, and David M. Kreps, 1979. Martingales and arbitrage in multiperiod securities markets. *Journal of Economic Theory* 20, 381–408.
- Hasbrouck, Joel, and Gideon Saar, 2013. Low-latency trading, *Journal of Financial Markets* 16, 646–679.
- Hendershott, Terrence, Charles M. Jones, C. M. and Albert J. Menkveld, 2011. Does algorithmic trading improve liquidity? *Journal of Finance* 66 1–33.
- Hendershott, Terrence, and Ryan Riordan, 2009. Algorithmic trading and information. *Working paper*, University of California, Berkeley.
- Jovanovic, Boyan, and Albert J. Menkveld, 2011. Middlemen in limit-order markets. *Working paper*, New York University.
- Krauzlis, Richard J. and Steve G. Lisberger, 1994. Temporal properties of visual motion signals for the initiation of smooth pursuit eye movements in monkeys. *Journal of Neurophysiology* 72, 150–162.
- Laughlin, Gregory, Anthony Aguirre, and Joseph Grundfest, 2014. Information transmission between financial markets in Chicago and New York. *The Financial Review*, this issue.
- Madhavan, Ananth, N. 2000. Market microstructure: A survey. *Journal of Financial Markets* 3, 205–258.

- McInish, Thomas H., and James Upson, 2013. The quote exception rule: Giving high frequency traders an unintended advantage. *Financial Management* 42, 481–501.
- Menkveld, Albert J., 2013. High frequency trading and the *new market* makers. *Journal of Financial Markets* 16, 712–740.
- Menkveld, Albert J., and Bart Z. Yueshen, 2012. Middlemen interaction and its effect on market quality. *Working paper*, VU University Amsterdam.
- O'Hara, Maureen 1995. *Market Microstructure Theory*. (Blackwell Publishers, Malden, MA).
- O'Hara, Maureen, and Mao Ye, 2011. Is market fragmentation harming market quality? *Journal of Financial Economics* 100, 459–474.
- Overdahl, James, and Henry McMillan, 1998. Another day, another collar: An evaluation of the effects of NYSE rule 80a on trading costs and intermarket arbitrage. *Journal of Business* 71, 27–53.
- Patterson, Scott, and Andrew Ackerman, 2012. SEC may ticket speeding traders, *Wall Street Journal*, February 23.
- Patterson, Scott, Jenny Strasburg, and Liam Plevin, 2013. High-speed traders exploit loophole, *Wall Street Journal*, May 1.
- SEC, 2010a. Concept release on equity market structure. Release No. 34-61358; File No. S7-02-10.
- SEC, 2010b. Findings regarding the market events of May 6, 2010. Report of the staffs of the CFTC and SEC to the joint advisory committee on emerging regulatory issues.
- Stoll, Hans R., 2001. Market fragmentation. *Financial Analysts Journal* 57, 16–21.

Panel A: Economics of securities trading



Panel B: Institutional structure

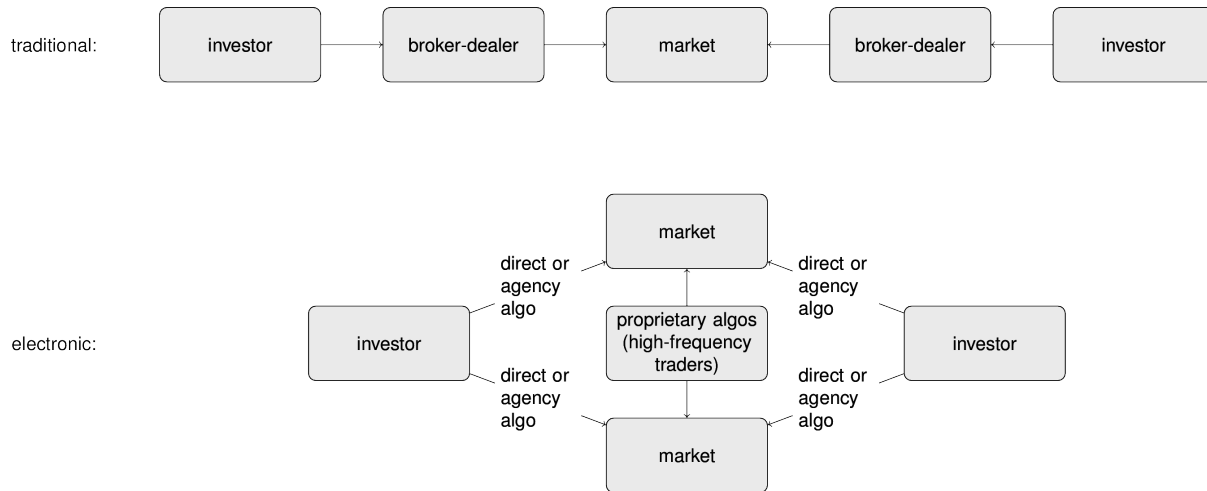


Figure 1

Traditional markets versus electronic markets

This figure illustrates, in the most parsimonious way, the economics of trading (panel A) and the difference between traditional markets and electronic markets (panel B).

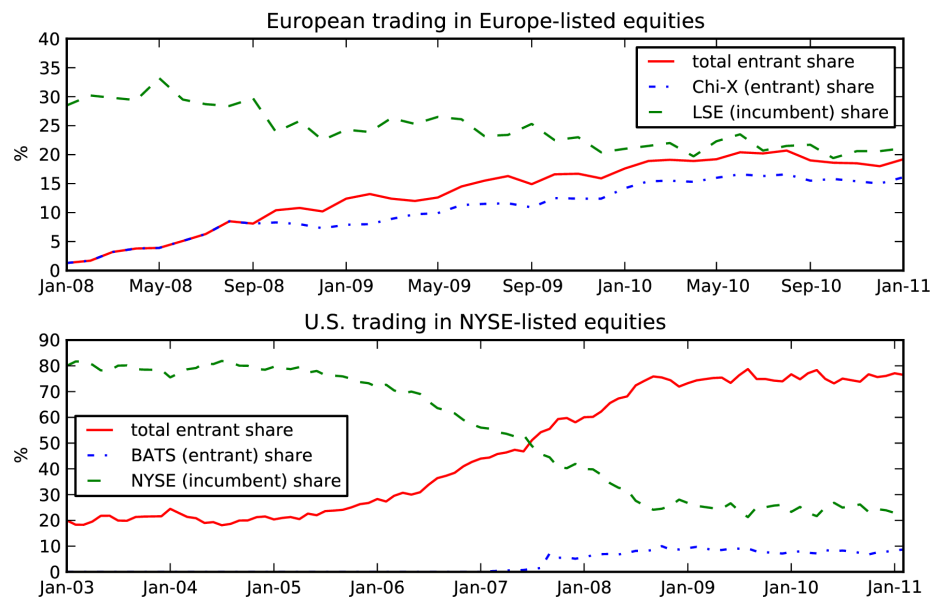


Figure 2

Market fragmentation

These graphs illustrate the entry of new markets. The top graph presents the total entrants' volume share in Europe-listed stocks as well as the largest entrant share (Chi-X) and the largest incumbent share (London Stock Exchange). The bottom graph presents the total entrants' volume share in NYSE-listed stocks as well as a large entrant share (BATS) and the incumbent share (NYSE). Source: Barclays Capital Equity Research and the Federation of European Securities Exchanges.

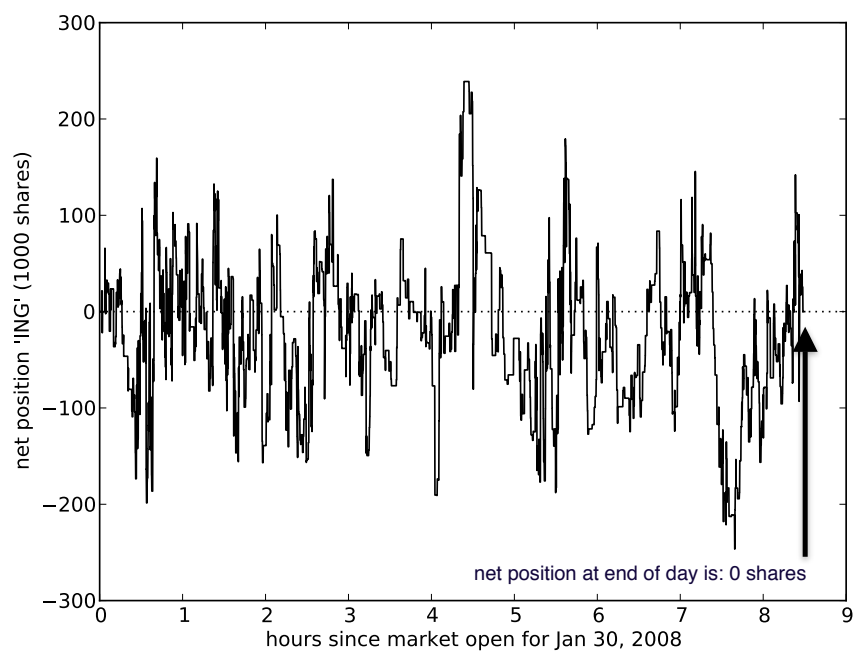


Figure 3

One day in the life of a high-frequency trader

This graph illustrates the nature of high frequency trading. It depicts the position of one broker in ING stock on January 30, 2008. The position is the aggregate net position across the incumbent market, Euronext, and an entrant market, Chi-X. Source: Jovanovic and Menkveld (2011).

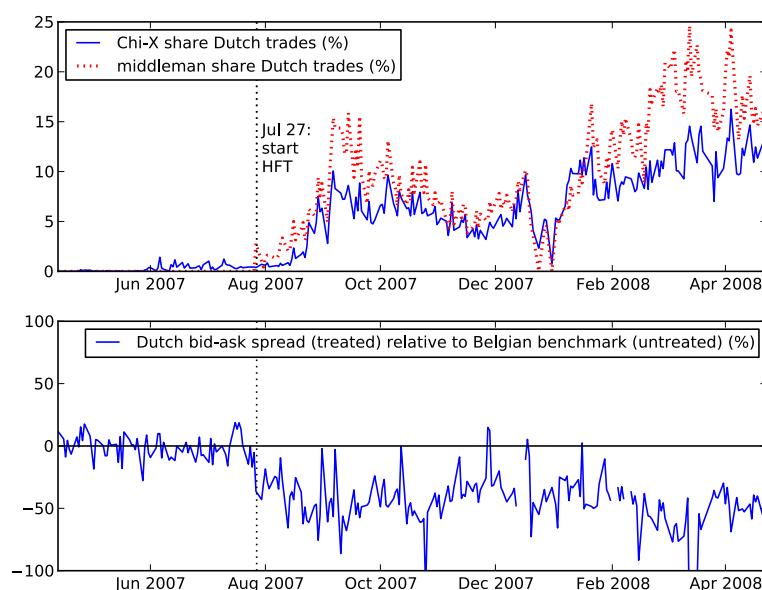


Figure 4

High-frequency trader activity, entrant market share, and the bid-ask spread

This figure depicts the evolution of trading in Dutch index stocks in the incumbent market Euronext and the entrant market Chi-X. The top graph plots the market share of Chi-X along with the participation rate (in both markets) of a new high-frequency trader. The bottom graph plots the average bid-ask spread of the ‘treated’ Dutch stocks relative to ‘untreated’ Belgian benchmark stocks (that did not experience Chi-X entry, nor entry of the new HFT). Source: Menkveld (2013).